

~\Desktop\LAB\queue.c

```
1  #include<stdio.h>
2  int A[20],size,front,rear;
3
4  // Function for inserting an item to Queue
5  void ENQUEUE(int item)
6  {
7      if(rear==size-1)
8          printf("Queue overflow\n");
9      else if(rear==-1)
10     {
11         front=0;
12         rear=0;
13         A[rear]=item;
14     }
15     else
16     {
17         rear++;
18         A[rear]=item;
19     }
20 }
21
22 // Function for deleting an item from Queue
23
24 void DEQUEUE()
25 {
26     if(front==-1)
27         printf("Queue is empty\n");
28     else if(front==rear)
29     {
30         printf("Deleted item is %d\n",A[front]);
31         rear=-1;
32         front=-1;
33     }
34     else
35     {
36         printf("Deleted item is %d\n",A[front]);
37         front++;
38     }
39 }
40
41 // Function for displaying the items in the Queue
42
43 void display()
44 {
45     int i;
46     if(front==-1)
47     {
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48         printf("Queue is empty\n");
49     }
50     else
51     {
52         printf("The queue elements are \n");
53         printf("f->");
54         for(i=front;i<=rear;i++)
55         {
56             printf("%d\t",A[i]);
57         }
58         printf("<-r\n");
59         printf("front=%d\n",front);
60         printf("rear=%d\n",rear);
61     }
62 }
63 }
64
65 // Function for main()
66
67 int main()
68 {
69     int opt,item;
70     rear=-1;
71     front=-1;
72     printf("Enter the size of Queue: \n");
73     scanf("%d",&size);
74     do
75     {
76         printf("Menu\n");
77         printf("\n 1.ENQUEUE\t 2.DEQUEUE \t 3.DISPLAY \t 4.EXIT");
78         printf("\n Enter the option:");
79         scanf("%d",&opt);
80         switch(opt)
81         {
82             case 1 : printf("Enter the item to be inserted: \n");
83                       scanf("%d",&item);
84                       ENQUEUE(item);
85                       display();
86                       break;
87             case 2 : DEQUEUE();
88                       display();
89                       break;
90             case 3 : display();
91                       break;
92             case 4 : break;
93
94             case 5 : printf(" \nInvaild choice!Please try again");
95         }
96     }while(opt!=4);
97 }
```

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98 OUTPUT
99
100 Enter the size of Queue:
101 3
102 Menu
103
104 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
105 Enter the option:2
106 Queue is empty
107 Menu
108
109 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
110 Enter the option:1
111 Enter the item to be inserted:
112 3
113 The queue elements are
114 f->3 <-r
115 front=0
116 rear=0
117 Menu
118
119 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
120 Enter the option:1
121 Enter the item to be inserted:
122 7
123 The queue elements are
124 f->3 7 <-r
125 front=0
126 rear=1
127 Menu
128
129 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
130 Enter the option:1
131 Enter the item to be inserted:
132 9
133 The queue elements are
134 f->3 7 9 <-r
135 front=0
136 rear=2
137 Menu
138
139 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
140 Enter the option:1
141 Enter the item to be inserted:
142 8
143 Queue overflow
144 The queue elements are
145 f->3 7 9 <-r
146 front=0
147 rear=2

```

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148 Menu
149
150 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
151 Enter the option:2
152 Deleted item is 3
153 The queue elements are
154 f->7 9 <-r
155 front=1
156 rear=2
157 Menu
158
159 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
160 Enter the option:2
161 Deleted item is 7
162 The queue elements are
163 f->9 <-r
164 front=2
165 rear=2
166 Menu
167
168 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
169 Enter the option:1
170 Enter the item to be inserted:
171 4
172 Queue overflow
173 The queue elements are
174 f->9 <-r
175 front=2
176 rear=2
177 Menu
178
179 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
180 Enter the option:2
181 Deleted item is 9
182 Queue is empty
183 Menu
184
185 1.ENQUEUE 2.DEQUEUE 3.DISPLAY 4.EXIT
186 Enter the option:4
187
188

```